	Case Name: Residential House (1)	Sector	Construction (residential building)	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
			One of nine std. scenarios	
Submitted by	William & Jef		User defined distributions	
Date	2016	Project Control	Automatic tracking	
File Name	C2016-11 Residential House (1)		Tracking based on user input	

1. Project description

Project authenticity

The construction of a residential house from foundation to completion. The project consists of activities such as site preparation, structural works, installation of utilities, and finishing, with associated resource and cost data obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	64
Planned Duration (PD)	241
Budget At Completion (BAC)	162,472.00 €
Renewable Resources	-
Consumable Resources	-

Network topology	
Serial/Parallel (SP)	57%
Activity Distribution (AD)	77%
Length of Arcs (LA)	52%
Topological Float (TF)	16%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0.00	0.00	N/A
CRI-rho	100	0.00	N/A
CRI-tau	100	0.00	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	58	49	-0.35
SI	13	26	2.44
SSI	8	11	2.10
CRI-r	12	12	1.80
CRI-rho	12	12	1.79
CRI-tau	10	9	1.25

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section "2.2. Risk Analysis". Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	1	0.00
PV - SPI	2	0.00
PV - SCI	2	0.00
ED - 1	1	0.00
ED - SPI	2	0.00
ED - SCI	2	0.00
ES - 1	2	-1
ES - SPI(t)	8	-6
ES - SCI(t)	8	-6

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.00	0.00
CPI	0.00	0.00
SPI	0.00	0.00
SPI(t)	1	0.00
SCI	6	-5
SCI(t)	1	0.00
0.8 CPI + 0.2 SPI	6	-5
0.8 CPI + 0.2 SPI(t)	0.00	0.00

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 5 tracking periods. The Real Duration and Real Cost mentioned in section "2.3.3. Earned Value Management" are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-5.40	-8160.80	-6.4	1.00	0.87	0.95	1.00
std dev	720.95	5233.01	3.78	0.01	0.13	0.02	0.00
final	-717.00	0.00	-13	1.00	1.00	0.95	1.00

2.3.3.2. Time forecasting

PD	241
Real Duration	254
Late/early	**5.39%**

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	253.40	7.83	2.13	0.24
PV - SPI	282.20	51.72	13.15	-11.10
PV - SCI	282.00	50.68	12.76	-11.02
ED - 1	255.40	2.70	0.87	-0.55
ED - SPI	284.80	49.41	12.13	-12.13
ED - SCI	284.40	49.11	11.97	-11.97
ES - 1	247.40	3.78	2.60	2.60
ES - SPI(t)	253.20	5.22	1.57	0.31
ES - SCI(t)	252.80	5.02	1.57	0.47

2.3.3.3. Cost forecasting

BAC	162,472.00 €
Real Cost	163,189€
Over/under budget	**0.44%**

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	162,477.40	806.05	0.49	0.44
CPI	162,270.65	1,326.48	0.79	0.56
SPI	181,946.54	30,908.45	11.49	-11.49
SPI(t)	166,318.00	4,581.05	1.93	-1.92
SCI	181,655.84	30,449.17	11.32	-11.32
SCI(t)	166,096.28	4,367.95	1.90	-1.78
0.8 CPI + 0.2 SPI	165,207.14	4,042.07	1.60	-1.24
0.8 CPI + 0.2 SPI(t)	163,036.16	1,250.35	0.59	0.09